

**A PROJECT REPORT ON**

# **COURIER MANAGEMENT SYSTEM**

*Developed on the Salesforce Developer Edition Platform*

Submitted in partial fulfilment of the requirements for the award of the degree of

**department of data science**

Submitted by

S.Gomathy

R. Deepika

K.Gayathri

P.gokula sundhari

R.Dhivya

**KAMBAN COLLEGE OF ARTS AND SCIENCE**

DEPARTMENT OF data science

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**BONAFIDE CERTIFICATE**

This is to certify that the project report entitled "COURIER MANAGEMENT SYSTEM" is a bonafide record of work done by group project in partial fulfilment of the requirements for the award of the degree of Bachelor of data science during the academic year 2025 - 2026.

The project has been carried out under my supervision and guidance, and it has not been submitted elsewhere for the award of any other degree or diploma.

Project Guide

Head of the Department

Submitted for the Viva-Voce examination held on \_\_\_\_\_

Internal Examiner

External Examiner

# ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported me throughout this project. I am extremely thankful to my project guide, Princy Jasmine, for her valuable guidance, constant encouragement, and constructive suggestions that helped me complete this project successfully.

I would also like to thank the Head of the Department and all the faculty members of the Department of Computer Applications, Kamban College of Arts and Science, for

providing the necessary facilities and a conducive environment to carry out this project.

I sincerely thank Salesforce for providing a free Developer Edition environment, which made it possible for me to design, build and test the Courier Management System without any cost, and to gain practical exposure to a real-world cloud platform.

Finally, I would like to thank my family and friends, especially my teammate Harshitha, for their constant support and encouragement throughout this project.

# ABSTRACT

Courier and parcel delivery companies handle a large volume of day-to-day operations such as booking new shipments, assigning couriers to delivery staff, tracking the status of a parcel, managing customer details, and generating delivery reports. When these activities are carried out manually or through scattered spreadsheets, they become time-consuming, error-prone, and difficult to scale as the business grows.

The “Courier Management System” is a cloud-based application developed on the Salesforce Developer Edition platform to digitise and automate these day-to-day courier operations. The system allows an organisation to register customers, create and track courier/parcel bookings, assign delivery staff, update delivery status in real time, and maintain a complete history of every shipment inside a single, centralised and secure database.

The application is built entirely using Salesforce’s declarative, point-and-click tools. Custom Objects were created to model real-world business entities such as Customer, Courier Booking, Delivery Staff and Branch. Custom Tabs were configured to give users an intuitive way to navigate between these objects, and the Lightning App Builder was used to assemble everything into a single, easy-to-use application.

This report documents the complete life cycle of the project — starting from signing up for a free Salesforce Developer Edition account, through system analysis, requirement gathering, design of the data model, step-by-step implementation inside Salesforce Setup (illustrated with actual screenshots captured during development), testing, and finally the conclusion and scope for future enhancement. The project demonstrates how a modern cloud platform can be used to build a functional, scalable courier management solution with minimal code and a short development time.

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## **CHAPTER 1 – INTRODUCTION**

### **1.1 Introduction**

The courier and logistics industry forms the backbone of e-commerce and business-to-business trade, moving millions of parcels every day between customers, warehouses and delivery destinations. As the volume of shipments grows, courier companies find it increasingly difficult to manage bookings, track deliveries and maintain customer records using manual registers or basic spreadsheets.

The Courier Management System is a web-based application designed to overcome these difficulties by providing a single, centralised platform to manage every stage of the courier process — from the moment a customer books a parcel to the moment it is delivered. The system is built on Salesforce, a leading cloud Customer Relationship Management (CRM) platform, using its Developer Edition, which is provided free of cost for learning and building applications.

Salesforce was chosen for this project because it allows an application to be built rapidly using declarative (no-code / low-code) tools such as Custom Objects, Custom Tabs, Page Layouts and the Lightning App Builder, while still offering the security, scalability and reliability of an enterprise-grade cloud platform.

### **1.2 Problem Statement**

Small and medium-sized courier businesses commonly rely on manual registers, paper receipts or disconnected Excel sheets to record parcel bookings and delivery status. This approach leads to several recurring problems:

- Duplicate or missing customer and parcel records due to manual data entry.
- No real-time visibility into where a parcel is or who is handling it.
- Difficulty in generating reports on daily bookings, pending deliveries or revenue.
- Inefficient assignment of delivery staff, leading to delays.
- Data loss risk since records are not backed up in a central, secure database.

There is, therefore, a clear need for a centralised, automated Courier Management System that can store all information securely in the cloud, be accessed from anywhere, and provide accurate, real-time status of every booking.

## 1.3 Objectives

The main objectives of this project are:

To design and implement a centralised database to store customer, booking, and delivery-staff information.

To provide an easy-to-use interface for creating and tracking courier bookings from pickup to delivery.

To reduce manual paperwork and human error in day-to-day courier operations.

To enable quick assignment of delivery staff to a booking and update its delivery status.

To make use of Salesforce's built-in reporting tools to generate business insights.

To build the entire application using declarative Salesforce tools, keeping development time and cost low.

## ● 1.4 Scope of the Project

- The scope of the Courier Management System covers the core operational needs of a small to medium courier business. It includes maintaining customer master data, recording courier/parcel bookings with sender and receiver details, assigning delivery staff, updating and tracking delivery status, and viewing basic reports. The system is developed entirely on the Salesforce Developer Edition environment and is intended as an academic / proof-of-concept project; it can be extended further with additional automation such as Apex triggers, Flows, email notifications and a customer-facing tracking portal, as discussed in Chapter 7.
- 

## CHAPTER 2 – SYSTEM ANALYSIS

### 2.1 Existing System

In the existing (manual) system followed by many small courier operators, parcel details are written down in physical registers or maintained in isolated spreadsheet files on individual computers. Customer details are collected separately at the counter, and there is no direct link between a customer record and their booking history.

Tracking the status of a parcel typically requires a phone call to the branch or the delivery staff, since there is no central system that everyone can refer to. Reports on how many parcels were booked, delivered or are pending have to be compiled manually at the end of the day, which is time-consuming and prone to mistakes.

Drawbacks of the existing system include:

Time-consuming manual record keeping.

High chance of data duplication and human error.

No centralised access – records are scattered across registers/files.

Difficult and slow report generation.

No real-time tracking of parcel/delivery status.

- 
-

## 2.2 Proposed System

- The proposed Courier Management System replaces manual registers with a single cloud-based application built on Salesforce. All customer, booking, and delivery information is stored in Custom
- Objects inside a secure Salesforce organisation ("org"), which can be accessed by authorised staff
- from any device with an internet connection.

Advantages of the proposed system:

Centralised, secure, cloud-hosted database – no data silos.

Quick creation and retrieval of customer and booking records.

Real-time status updates visible to all authorised users.

Built-in Salesforce reports and dashboards for instant business insight.

Role-based access control ensures data security.

● Scalable – new objects, fields, or automation can be added as the business grows.

## ● 2.3 Feasibility Study

- A feasibility study was carried out before development to ensure that the project could be
- completed successfully within the available time and resources.

### ● 2.3.1 Technical Feasibility

Salesforce Developer Edition provides all the tools required to build this project — Setup, Object Manager, Custom Tabs, and the Lightning App Builder — without needing any additional software or infrastructure. This made the project technically feasible with the skills acquired during the course.

### 2.3.2 Economic Feasibility

Salesforce Developer Edition is completely free of cost, which means the project required no financial investment in licences, servers, or hosting, making it highly economical for an academic project.

### 2.3.3 Operational Feasibility

The declarative, point-and-click nature of Salesforce made it easy to design and operate the system even without writing extensive code, making the solution practical for day-to-day use by non-technical staff after basic training.

## CHAPTER 3 – SYSTEM REQUIREMENTS

### 3.1 Hardware Requirements

| Component           | Specification                                                    |
|---------------------|------------------------------------------------------------------|
| Processor           | Intel i3 / AMD equivalent or higher                              |
| RAM                 | 4 GB minimum (8 GB recommended)                                  |
| Hard Disk           | 20 GB free space                                                 |
| Internet Connection | Broadband connection (required, since Salesforce is cloud-based) |
| Display             | Standard laptop / desktop monitor                                |

### 3.2 Software Requirements

| Component        | Specification                                     |
|------------------|---------------------------------------------------|
| Operating System | Windows 10 / 11 (or any OS with a modern browser) |
| Platform         | Salesforce Developer Edition (cloud-based, free)  |
| Browser          | Google Chrome (recommended), Microsoft Edge       |
| Email Client     | Gmail (for account verification / password        |

|                     |                                      |
|---------------------|--------------------------------------|
|                     | reset)                               |
| Documentation Tools | Microsoft Word, Microsoft PowerPoint |

## CHAPTER 4 – SYSTEM DESIGN

### 4.1 System Architecture

The Courier Management System follows the standard three-tier architecture provided by the Salesforce platform:

- Presentation Layer – Salesforce Lightning Experience, accessed through a web browser, where users interact with Tabs, Page Layouts and the App Launcher.
- Application / Logic Layer – Salesforce’s built-in platform services, including declarative logic (validation rules, page layouts) that govern how data is captured and displayed.
- Data Layer – Custom Objects stored securely in the multi-tenant Salesforce database (Force.com database), including Customer, Courier Booking and Delivery Staff records.

Because Salesforce is a Platform-as-a-Service (PaaS), the underlying servers, database engine, security patches and backups are all managed by Salesforce, allowing the project to focus purely on designing the business application itself.

### 4.2 Data Flow / Process Diagram

The overall process flow of the system can be summarised as follows:

- Step 1: Customer approaches the courier branch and provides sender / receiver details.
- Step 2: Staff creates a new Customer record (if not already existing) in the system.
- Step 3: Staff creates a new Courier Booking record, linking it to the Customer.
- Step 4: A Delivery Staff member is assigned to the booking.
- Step 5: The booking status is updated as it moves through stages – Booked → Picked Up → In Transit → Delivered.
- Step 6: Reports are generated from the stored data for business review.

### 4.3 Module Description

The system is divided into the following functional modules:

#### 4.3.1 Customer Management Module

Handles the creation, updating and searching of customer records, including name, contact number, and address details of both sender and receiver.

#### 4.3.2 Courier Booking Module

Handles the creation of a new courier/parcel booking, capturing pickup and delivery address, parcel weight/type, booking date and linking the booking to a customer record.

#### 4.3.3 Delivery Management Module

Handles the assignment of delivery staff to a booking and the tracking of delivery status from pickup to final delivery.

#### 4.3.4 Reports Module

Uses Salesforce's native Reports and Dashboards to display information such as number of bookings per day, pending deliveries, and completed deliveries.

### 4.4 Data Model (Custom Objects)

The following Custom Objects were designed in the Salesforce Object Manager to represent the core entities of the Courier Management System:

| Custom Object   | Key Fields                                                                   | Purpose                              |
|-----------------|------------------------------------------------------------------------------|--------------------------------------|
| Customer        | Name, Phone, Email, Address                                                  | Stores sender/receiver master data   |
| Courier Booking | Booking No, Customer, Pickup Address, Delivery Address, Status, Booking Date | Stores each parcel booking           |
| Delivery Staff  | Staff Name, Contact No, Assigned Bookings                                    | Stores delivery personnel details    |
| Branch          | Branch Name, Location                                                        | Stores courier branch/office details |

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

## **CHAPTER 5 – IMPLEMENTATION**

### **5.1 Tools and Technology Used**

- Salesforce Developer Edition – free cloud CRM/PaaS platform used to build the entire application.
- Salesforce Setup – the administration console used to create Custom Objects, Custom Tabs and configure the org.
- Lightning Experience – the modern Salesforce user interface used to run and test the application.
- Object Manager – used to design the data model (Custom Objects and fields).
- Google Chrome – web browser used to access Salesforce.
- Gmail – used to receive the account confirmation and password-reset e-mail sent by Salesforce during sign-up.

### **5.2 Step-by-Step Implementation**

This section documents, with screenshots taken directly during development, the actual steps followed to build the Courier Management System on Salesforce.

#### **Step 1: Signing up for a Salesforce Developer Edition Account**

The first step in building the Courier Management System was to create a free Salesforce Developer Edition account. The sign-up form was filled in with the developer's first name, last name, job title, company name (Kamban College of Arts and Science), country and a work e-mail address. Salesforce validates each field in real time, shown by the green tick marks next to every completed field.

The image shows a sign-up form for Salesforce Developer Edition. The form is titled "Sign up for your Developer Edition" and includes a sub-header "A free Salesforce Platform environment with Agentforce and Data 360". The form fields are: First name (Dheepika), Last name (R), Job title (Developer), Company (Kamban college of arts and...), Country/region (India), and Work email (rdheepika04@gmail.com). There is a checkbox for agreeing to the terms of service and a phone icon next to the email field.

**Sign up for your Developer Edition**

A free Salesforce Platform environment with Agentforce and Data 360

First name  
Dheepika ✓

Last name  
R ✓

Job title  
Developer ✓

Company  
Kamban college of arts and... ✓

Country/region  
India ✓ ▼

Work email  
rdheepika04@gmail.com 

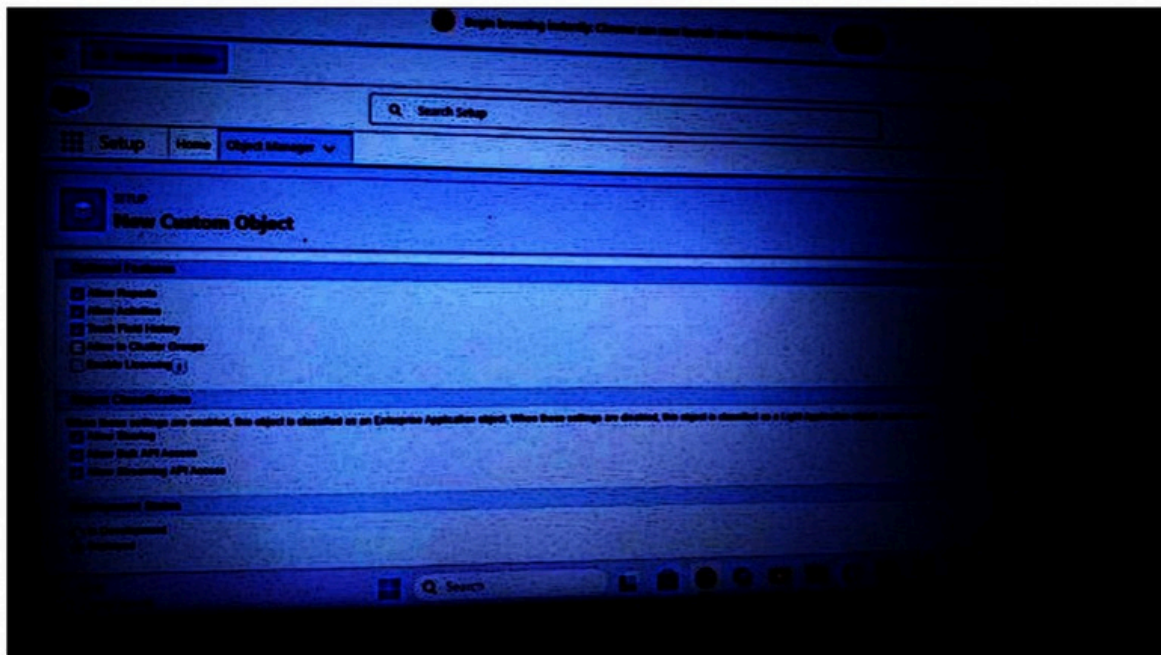
☐ I agree to the [Main Services Agreement - Developer Services](#), [Salesforce Program Agreement](#), and [API Terms of Service](#)

Your free trial may be provisioned on or

*Fig 5.1 – Salesforce Developer Edition sign-up form*

## **Step 2: Verifying the Account through E-mail**

After submitting the sign-up form, Salesforce sends a confirmation e-mail titled "Welcome to Salesforce: Reset your password" to the registered Gmail address. This e-mail contains a link that is used to set a password and activate the new Developer Edition org for the first time.



*Fig 5.6 – Configuring Optional Features and Deployment Status for a new Custom Object*

### **Step 7: Creating Custom Tabs for Navigation**

After the Custom Objects were created, Custom Tabs were configured from Setup → User Interface → Tabs so that users could navigate to each object directly from the app navigation bar. The Custom Tabs page (shown below) lists the different tab types supported by Salesforce — Custom Object Tabs, Web Tabs, Visualforce Tabs, Lightning Component Tabs and Lightning Page Tabs — and provides a “New” button to create one for each Custom Object, such as Courier Booking and Delivery Staff.



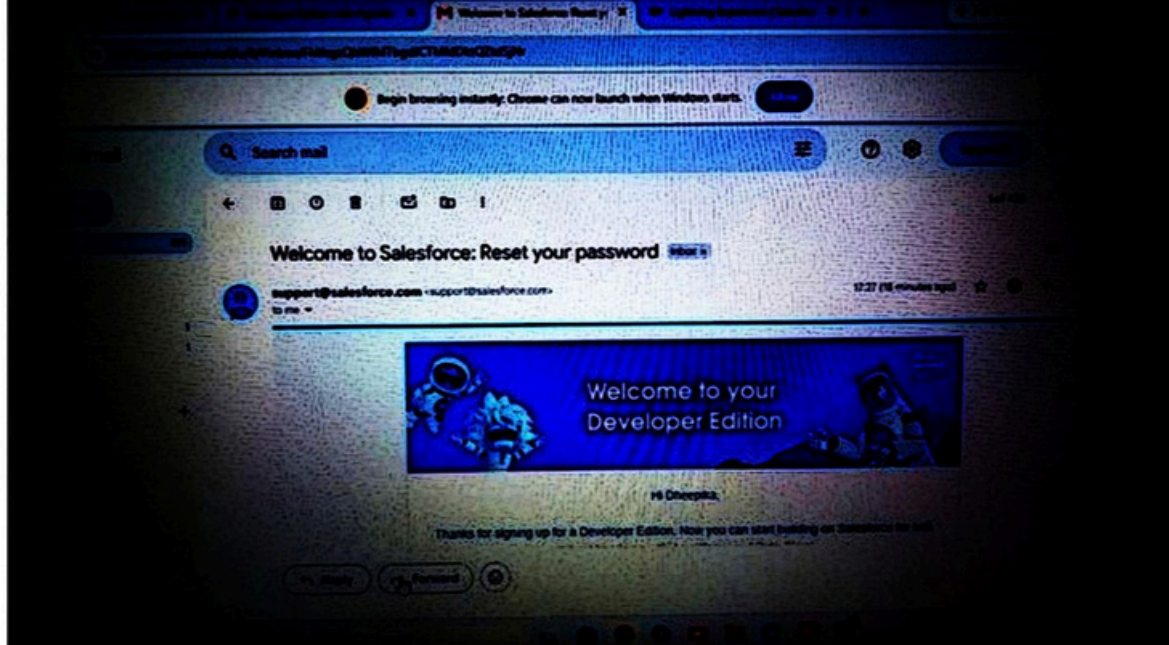
| 1 | Create a new Customer record                      | Record saved and visible in Customer tab      | Pass |
|---|---------------------------------------------------|-----------------------------------------------|------|
| 2 | Create a new Courier Booking linked to a Customer | Booking saved with correct customer reference | Pass |
| 3 | Assign Delivery Staff to a Booking                | Staff field updated on the Booking record     | Pass |
| 4 | Update Booking status to 'Delivered'              | Status field reflects updated value           | Pass |
| 5 | Open each Custom Tab from the navigation bar      | Correct object list view opens                | Pass |
| 6 | Generate a report of all bookings                 | Report displays all Courier Booking records   | Pass |

## CHAPTER 7 – CONCLUSION AND FUTURE ENHANCEMENT

### 7.1 Conclusion

The Courier Management System successfully demonstrates how a cloud platform such as Salesforce can be used to design and build a fully functional business application without writing large amounts of code. By creating Custom Objects for Customer, Courier Booking, Delivery Staff and Branch, and configuring Custom Tabs for easy navigation, the project provides a centralised system to manage the complete lifecycle of a courier booking — from customer registration to final delivery.

The project also gave practical, hands-on experience of real-world software development activities such as requirement analysis, feasibility study, data-model design, declarative configuration inside Salesforce Setup, and testing — all carried out on a genuine, industry-used cloud platform (Salesforce Developer Edition) rather than a simulated environment.



*Fig 5.2 – Account verification e-mail received in Gmail*

### **Step 3: Logging in and Landing on the Setup Home Page**

Once the password was set, logging in for the first time takes the user directly to the Salesforce Setup Home page. This page, shown below, displays a welcome message with the developer's name ("Welcome, Dheepika"), confirming that the org has been created successfully, along with quick links to commonly used setup areas such as Users, Release Updates and the Salesforce Mobile App.

**Begin browsing instantly:** Chrome can now launch when Windows starts.

**CD Developer Edition**

### Q Search Setup

**Setup**

## 1. Introduction

Object Manager v

**Q Quick Links**

Welcome, Dheepika

Manage and customize Salesforce from Setup. Browse suggestions, explore features, and more.

**Get Answers and Complete Setup Tasks Faster with an Agent**

Use Setup with Agentforce to ask questions, get step-by-step guidance, or have the agent complete tasks for you, all without searching through docs or pages.

## Review and configure settings to get started

Configure

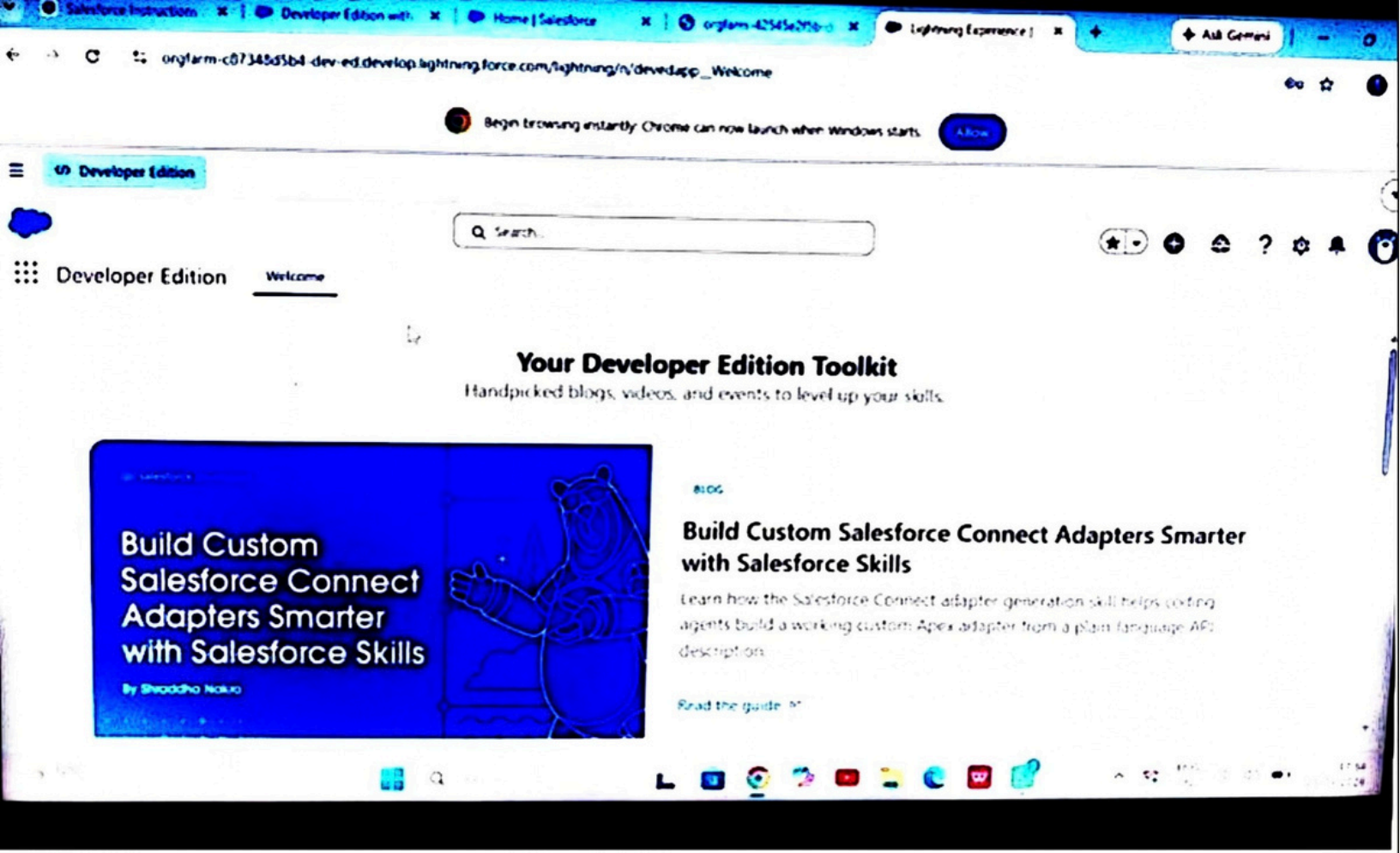
2005

## Achieve Popular Business Goals

— 9 —

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Begin browsing instantly. Chrome can now launch when Windows starts.

Allow



Developer Edition



Search



Developer Edition

Welcome

## Your Developer Edition Toolkit

Handpicked blogs, videos, and events to level up your skills.



BLOG

### Build Custom Salesforce Connect Adapters Smarter with Salesforce Skills

Learn how the Salesforce Connect adapter generation skill helps coding agents build a working custom Apex adapter from a plain language API description.

[Read the guide](#)

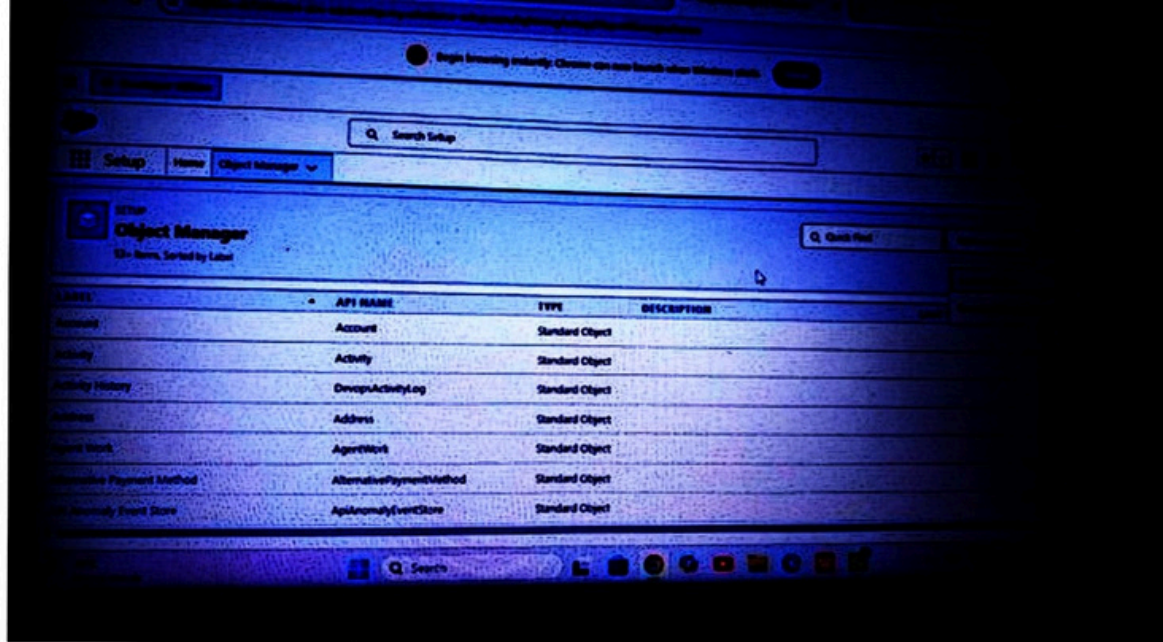
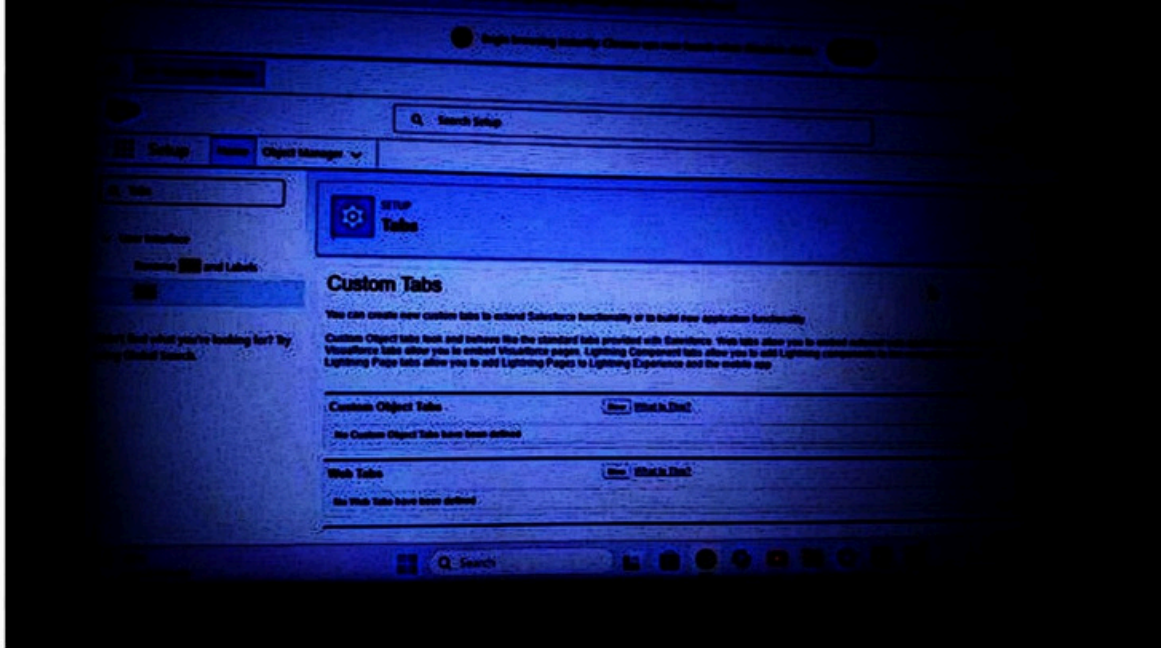


Fig 5.5 – Object Manager with the 'Create > Custom Object' option

### Step 6: Configuring a New Custom Object

On the "New Custom Object" page, the required Optional Features were enabled — Allow Reports and Track Field History — so that data captured in the object (for example, courier bookings) can later be summarised in Salesforce reports and any change to a record can be audited. Under Object Classification, Allow Sharing, Allow Bulk API Access and Allow Streaming API Access were also enabled so the object behaves as a full Enterprise Application object. Finally, the Deployment Status was set to "Deployed" so that the object becomes immediately available to users, instead of remaining hidden "In Development."



*Fig 5.7 – Setup > Tabs page used to create Custom Object Tabs*

### **Step 8: Final Verification of the Configured Org**

After the Custom Objects, fields, and Tabs were created and deployed, the Setup Home page was revisited to verify that the configuration had been saved correctly and that the org ("Welcome, Dheepika") was fully set up and ready for the Courier Management System app to be assembled using the Lightning App Builder.

# SYSTEM (SECMS)

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## A SALESFORCE PROJECT

Submitted to

KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN

Department of Data Science

Academic Year 2024-2027

## PROJECT TEAM DETAILS

Team Members:

GOMATHY S - gomathysampath2006@gmail.com

GAYATHRI.K - gayu25122007@gmail.com

DHEEPIKA. - rrdheepika04@gmail.com

DHIVIYA R - ragothamandhivya@gmail.com

GOKULASUNDARI P - poomalaimangai@gmail.com

## Guided By:

Department of Data Science

Kamban College of Arts and Science for Women

